

Analyzing volatility spillovers and hedging in some agricultural commodity markets in India

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Abstract

Price volatility is a significant economic phenomenon and is endangering market participants to possibly excessive risk. This paper examines a market based solution, futures markets, and provides a complete analysis of the dynamics of volatility spillover across and within the thirteen agricultural commodity markets in India. In addition, this paper also evaluates the hedging effectiveness of agricultural commodity futures market. We employ a multivariate GARCH (asymmetric VAR BEKK) model to examine the significance of volatility spillovers and asymmetric impact of shocks on the conditional variance of futures and spot returns.

The estimation results indicate that there is a strong ARCH and GARCH effects in both spot and futures markets for almost all crops. There is a short term as well as long term volatility spill over across and within agricultural commodity markets. We see that agricultural commodity markets have turned out to be more interdependent in recent times. We calculated the optimal hedge ratios, optimal portfolio weights and hedge effectiveness index in order to provide important policy implications for risk management in agricultural commodity markets.

Key words: Futures market, Agricultural commodities; Volatility transmission, Hedging

JEL Classification: G13, Q11, C32 and G32

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1. Introduction

One of the major difficulties facing the producers of agricultural commodities is price uncertainty. The decisions regarding what to produce and how much to produce depends on the expected harvest price of the commodity; and given highly volatile supply and inelastic demand, the prices may not be the same as expected (Morgan 1999). The price volatility² impacts producers by reducing their expected income, which in turn affects their planning and investments. Government, then intervenes in the market to reduce the price uncertainty. However, government intervention put pressure on the exchequer (WORLD BANK 1996). This paper intends to examine a market based solution, futures markets³, and focuses on the modelling of agricultural commodity price volatility in a multivariate⁴ framework. It analyses the significance of volatility spillovers, and their impact on optimum hedging strategy⁵.

The volatility of agricultural commodity markets affects the economy at macro and micro level. There is a volatility spillover in and across agricultural commodity markets because the agricultural commodities are linked through substitutability and complementarity in demand and supply, share common market information, and are affected by herd behaviour and specific macroeconomic factors such as aggregate demand, inflation, interest rate etc. (Malliaris and Urrutia 1996). Therefore, understanding the spillover of price risks in and across the agricultural commodity markets are particularly significant for market participants, farmers and policy makers.

Since 1990s with economic reforms and WTO agreements, there has been a gradual opening-up of Indian agriculture to international markets. Although economic reforms generate new economic openings, it also throw up new challenges. Particularly,

² Volatility is defined as extent of price fluctuations around the trend.

³ How futures market reduce price uncertainty is discussed in detail in appendix A.

⁴ We use multivariate model because there is a high degree of correlation among the contemporaneous volatility shocks to agricultural commodity futures prices; and volatility shocks to one variable also gets transmitted to other variables.

⁵ Optimum hedging strategies improve the hedging role of futures markets and add to social welfare. Agricultural producers (market participants) improve their utility by reducing their portfolio variance (Bekarman 2011).

increased price volatility and the capacity to tackle it are again turning out to be most important policy concerns (WORLD BANK 1996).

One of the major reasons for establishing commodities futures markets in India in 2004 was to set up the infrastructure that would assist farmers to access the market as well-informed players. The National Agricultural Policy 2000 (NAP 2000) emphasized wider coverage of futures markets to minimize the fluctuations in commodity prices; and also for hedging price risk. The Guru committee (2001) laid emphasis on the role of futures trading for price risk management and marketing of agricultural commodities.

In agricultural markets, risks may arise due to drought, flood, high production of crops, unexpected jump in demand, a decline in global production, government policy, major regional or global economic crisis and geopolitical risks (Choudhry 2009). Market participants use hedging strategy to protect themselves from volatility. Futures market performs the role of risk transfer (Edward and Ma 2003, p. 422).

In view of the above discussion, the phenomenon⁶ of the volatility spillovers between futures and spot agricultural commodity markets are expected to be significant for forming hedge ratios and constructing optimal portfolios. In this wider context, this paper sets to examine the following objectives:

- a. To examine the significance of volatility spillovers and asymmetric⁷ impact of shocks on the conditional variance of futures and spot returns.
- b. To investigate the hedging effectiveness of agricultural commodity futures market.

⁶ A shock (news) in any market leads to the volatility in that market. There is volatility spill over from one market to other market because markets are linked. Market participants use hedging techniques to protect themselves from volatility. Market participants take opposite position in futures market to hedge. The knowledge about Volatility spill over is significant for hedging because it leads to co movement of the prices in both markets. If there is no volatility spill over from one market to other, hedging cannot be done (Chang, McAleer and Tansuchat 2011).

⁷ Asymmetry means bad news and good news (of equal magnitude) have different impact on the volatility of returns. Bad news may bring turmoil in commodity futures market, while good news may be less subverting.

In financial market literature the topic of volatility spillover and hedging has been extensively discussed. However, there are very few studies concerning agricultural commodity markets (Streeter and Tomek 1992; Apergis and Weaver 1999; Yang and Awokuse 2003; Buguk, Hudson, and Hanson 2003).

Choudhary (2009) analysed the hedging effectiveness of futures market in seven agricultural commodities using four different multivariate GARCH models, and found the time varying optimal hedge ratio. Beckerman (2011) incorporated market linkages and price of related commodities in hedge ratio estimation and analysed the possible gains in optimal hedge ratio calculation.

Previous studies have used multivariate Generalised autoregressive conditional heteroskedastic (GARCH) model models to estimate time varying hedge ratios for agricultural commodity markets (Choudhary 2009; Bekarman 2011). However, they did not examine the volatility spillover effects for each of the futures and spot agricultural commodity prices in and across all markets.

In addition, we note that even the latest research papers on this subject in the Indian context are more or less outdated, using data for the period immediately after setting up of three national exchanges, when the markets were not well established (Mahalik, Acharya and Babu 2009; Nath and Lingareddy 2008), and/or are very narrow in terms of crop coverage (Bhanumurthy, Dua and Kumawat 2013; Nath and Lingareddy 2008), and/or methodologically unsatisfactory (Shihabudheen and Padhi 2010) and/or analysed the spot and futures markets relationship only and did not examine the hedging strategies in Indian agricultural commodity markets (Sehgal, Ahmad and Deisting 2014).

The multivariate⁸ Generalised Autoregressive Conditional Heteroskedastic (GARCH) models are the most appropriate models for estimating OHRs as they show variability over time (Cecchetti *et al.* 1988 ; Baillie and Myers 1991; Kroner and Sultan 1993). There are several different formulations of multivariate GARCH models. We prefer Baba, Engle, Kraft and Kroner (BEKK) model because BEKK model makes sure that

⁸ We name the simultaneously modelling of shocks in two markets as multivariate GARCH model.

covariance matrix is positive semi definite and it gives information about the volatility spill over from one market to other market (Wang 2009, p.72).

This paper contributes to the literature in several ways. It employs more recent data for the period August 2009 to September 2014. It covers the major agricultural crops – namely, wheat, barley, maize, gram, mustard, castorseed, soyabean, chilli, coriander, pepper, cotton seed oilcake sugar and zeera. We were constrained in our choice of commodities by the fact that data were not available for other agricultural crops of interest.

Using daily data from August 2009 through September 2014 for these agricultural crops listed on the National Commodity and Derivative Exchange (NCDEX) and the Multi-Commodity Exchange (MCX), we find that the results provide a strong ARCH⁹ and GARCH¹⁰ effects in both spot and futures markets for almost all crops. There is a short term as well as long term volatility spill over across and within agricultural commodity markets. The evidence provided by optimal portfolio weights of all multivariate volatility models for all crops suggest that we should hold physical commodity in greater proportion than futures contracts.

The paper is organized as follows. Section 2 discusses the model used in the present study, and the estimation methodology adopted. Section 3 describes the economic implications for alternate hedging strategies. Section 4 presents a description of the data used. Section 5 presents the estimation results; and, finally, Section 6 provides the important conclusions and policy implications.

2. Model and methodology used in this study

To study the volatility spillover relationship between the spot and futures markets, we start off by measuring the volatility. There are two components of the price formation process: predictable and unpredictable components. We need to remove the predictable elements of the price process, as only unpredictable component

⁹ ARCH impact means that the past shocks in any market is affecting the present volatility of that market.

¹⁰ GARCH impact means that the past volatility in any market affecting the present volatility of that market

measures the volatility (Moledina, Roe and Shane 2003). Predictable elements include the elements of inflation, trend and seasonality. We remove the impact of inflation by deflating the spot and futures prices with the wholesale¹¹ price index. We then test for the stationarity of the spot and futures prices of each of our sample commodities. Since these were found to be nonstationary, we opted to work with the log (first) differences of the prices.

Further, we use the seasonal dummies to remove the impact of seasonality from the spot and futures prices. To determine if there are structural breaks, we conducted the Elliot and Muller (2006) test. Elliot and Muller (EM) test is improvement over other structural break tests because it allows for models with different types of structural breaks. There can be many or few breaks, clustered breaks, breaks that occur at regular intervals etc.

Volatility of returns signifies the risk in owning an asset. We, therefore, need to model volatility. We call innovations ε_t as “shocks” or “news” in the financial models, as innovations ε_t denote the unexpected, and we model the shock ε_t for modelling the volatility.

Spot prices are influenced by their past history, current and past futures prices, and other market information. Likewise, futures prices are affected by their past history, current and past spot prices and other market information (Kawaller, Koch and Koch 1987).

The following is the economic model of the study:

$$\begin{aligned} \text{A.} \quad d\ln SP_{ct} &= \alpha_1 d\ln FP_{c(t-1)} + \dots + \alpha_k d\ln FP_{c(t-k)} + \beta_1 d\ln SP_{c(t-1)} + \dots + \\ &\quad \beta_k d\ln SP_{c(t-k)} + \varepsilon_{1t} \end{aligned} \quad (1)$$

$$\begin{aligned} \text{B.} \quad d\ln FP_{ct} &= \gamma_1 d\ln FP_{c(t-1)} + \dots + \gamma_k d\ln FP_{c(t-k)} + \delta_1 d\ln SP_{c(t-1)} + \dots + \\ &\quad \delta_m d\ln SP_{c(t-k)} + \varepsilon_{2t} \end{aligned} \quad (2)$$

where c = wheat (W), maize (MZ), barley (BL), mustard (MS), gram (G), sugar (S), soyabean (SB), castorseed (CS), coriander (CD), pepper(PP), zeera (Z) and cotton

¹¹ We have taken the WPI of the individual crops to deflate the nominal prices.

seed oilcake(CO); SP is the spot price, FP is the futures price, and t denotes the time period, k is the optimal number of lags in the model.

We estimate¹² the economic model by vector auto regression (VAR). So our mean model is VAR. Equations (1) and (2) are the VAR model.

ε_{1t} is the shock in spot market one, and ε_{2t} is the shock in futures market. We assume that

$$\varepsilon_{1t}|\psi_{t-1} \sim N(0, h_{1t}) \quad (3)$$

$$\varepsilon_{2t}|\psi_{t-1} \sim N(0, h_{2t}), \quad (4)$$

Equations (3) and (4) say that ε_{1t} and ε_{2t} are normally distributed with mean zero and time varying variance¹³ h_{1t} and h_{2t} respectively, conditioned on the information available at time $t - 1$. ψ_{t-1} is the information available at time $t - 1$. Conditional variance h_{1t} implies volatility in spot returns and conditional variance h_{2t} implies volatility in futures returns.

We have simultaneously modelled the shocks in spot and futures returns. We employed Baba, Engle, Kraft and Kroner (BEKK) model because it makes sure that covariance matrix is positive semi definite, variance is positive, and it gives information about the volatility spill over from one market to other market (Wang 2009, p.72). Bad news and good news (of equal magnitude) have different impact on the volatility of returns. We, therefore employ asymmetric BEKK model in this study.

The BEKK model

If the number of assets are N, we model the variances and covariances of N assets.

The asymmetric BEKK model is written as:

$$H_t = W'W + B'H_{t-1}B + A'\varepsilon_{t-1}\varepsilon'_{t-1}A + D'z_{t-1}z'_{t-1}D \quad (5)$$

$$\varepsilon_t|\psi_{t-1} \sim N(0, H_t),$$

¹² We simultaneously estimate the mean and the variance of the series (Enders 2014, pp 124).

¹³ Variance of ε_{1t} is known as conditional variance of spot returns. Similarly, variance of ε_{2t} is known as the conditional variance of futures returns.

where H_t is a covariance matrix of order $N \times N$, ε_t is the innovation term of order $N \times 1$, ψ_{t-1} is the information available at time $t - 1$, A and B are unrestricted parameter matrix of order $N \times N$, C is symmetric parameter matrix of order $N \times N$, z_{t-1} is the column vector of order $N \times 1$.

$$z_{t-1} = \begin{cases} 0, & \varepsilon_{t-1} > 0 \\ -\varepsilon_{t-1}, & \varepsilon_{t-1} < 0 \end{cases} \quad (6)$$

If $N = 2$, then we have the following:

$$H_t = \begin{bmatrix} h_{11t} & h_{12t} \\ h_{21t} & h_{22t} \end{bmatrix}; C = \begin{bmatrix} c_{11} & c_{12} \\ c_{12} & c_{22} \end{bmatrix}; A = \begin{bmatrix} \alpha_{11} & \alpha_{12} \\ \alpha_{21} & \alpha_{22} \end{bmatrix}; B = \begin{bmatrix} \beta_{11} & \beta_{12} \\ \beta_{21} & \beta_{22} \end{bmatrix}$$

$$D = \begin{bmatrix} \delta_{11} & \delta_{12} \\ \delta_{12} & \delta_{22} \end{bmatrix}$$

3. The economic implications for alternate hedging strategies

One of the social benefits of futures market is the hedging. In this paper, we construct optimum portfolio design and hedging strategy by two methods. The first method is of Kroner and Ng (1998) where we calculate optimal portfolio weights, and the second approach is of Baillie and Myers (1991); Kroner and Sultan (1993) where we estimate time varying optimal hedge ratio by calculating multivariate GARCH models.

3.1. optimal portfolio weights

We can minimise risk without reducing expected returns by constructing an optimal portfolio design. We apply the techniques of Kroner and Ng (1998) to calculate the optimal portfolio weights. In the case of two assets (spot/futures), we calculate the portfolio weight as follows:

$$pw_{12,t} = \frac{h_{22,t} - h_{12,t}}{h_{11,t} - 2h_{12,t} + h_{22,t}} \quad (7)$$

Also,

$$pw_{12,t} = \begin{cases} 0, & \text{if } pw_{12,t} < 0 \\ pw_{12,t}, & \text{if } 0 \leq pw_{12,t} \leq 1 \\ 1, & \text{if } pw_{12,t} > 1 \end{cases} \quad (8)$$

Where $pw_{12,t}$ is the weight of first agricultural commodity (spot) in one rupee portfolio of two assets (spot and futures) at time t , $1 - pw_{12,t}$ is the weight of second agricultural commodity (futures) in one rupee portfolio of two assets (spot and futures) at time t . $h_{12,t}$ is the conditional covariance between agricultural commodities 1 (spot) and 2 futures, $h_{22,t}$ is the conditional variance of second (futures) agricultural commodity.

3.2. Optimum hedge ratio

We use futures market for hedging purpose. We hedge by choosing opposite positions in futures and spot market at the same time so as to offset the loss incurred from unfavourable price movement in one market by favourable price movement in other market.

We determine the dynamic hedge ratio (Baillie and Myers 1991; Kroner and Sultan 1993) in the following way:

$$\beta_t = \frac{h_{12,t}}{h_{22,t}} \quad (9)$$

where β_t is the risk minimising hedge ratio for two agricultural commodities 1 (spot) and 2 (futures), $h_{12,t}$ is the conditional covariance between two agricultural commodities 1 (spot) and 2 (futures), $h_{22,t}$ is the conditional variance of second commodity. So as to reduce risk, if we take a long (buy) position of one rupee in one agricultural commodity (spot) then we should hedge it by a short (sell) position of β_t rupees in another commodity (futures).

3.2.1. Hedging effectiveness index

We can compute the hedging effectiveness index (HEI) from the results obtained from multivariate GARCH model. We define the hedging effectiveness index¹⁴ (Chang et al. 2011) as follows:

$$HEI = \left[\frac{V_{unhedged} - V_{hedged}}{V_{unhedged}} \right] \quad (10)$$

¹⁴ The hedging effectiveness index is defined in terms of reduction in variance for a hedged portfolio when it is compared with unhedged portfolio.

Where HEI is the hedging effectiveness index, V_{hedged} is the variance of hedged portfolio. We obtain the variance of hedged portfolio from the variance of the return on portfolio¹⁵, i.e. $Var(Ret_{mt}|I_{t-1})$, $V_{unhedged}$ is the variance of unhedged portfolio which is the variance of spot returns. A high value of HEI implies high hedging effectiveness and greater reduction in risk.

4. The data set

In this study we use futures and spot price data for wheat (W), maize (MZ), barley (BL), mustard (MS), gram (G), sugar (S), soyabean (SB), castorseed (CS), coriander (CD), chilli (CH), pepper (PP), zeera (Z), and cotton seed oilcake (CO). Our choice of commodities was determined both by their importance as well by the simple fact of data availability. We realise that there are many other commodities of significance in India agriculture, but data were not available for them. The futures and spot price data were collected from the websites of the National Commodity and Derivative Exchange (NCDEX), Mumbai, and the Multi-Commodity Exchange (MCX), Mumbai. Alternative futures contracts are traded simultaneously on a daily basis. Of these, we chose the so-called 'nearby contract' data for our analysis, because it is the most liquid contract of all (Crain and Lee 1996). All data are available on a daily basis for six days a week (barring Sunday, and national holidays), from August 2009 through September 2014.

5. Estimation results

To study the volatility spillover relationship between the spot and futures markets, we start off by removing the impact of inflation, trend and seasonality of the spot and futures prices of each of our sample commodities. We use seasonal dummies to remove the seasonal effect. We found that there is seasonal impact in the spot price for maize, soyabean, wheat, zeera, and in futures price for coriander, cotton seed oil cake, mustard and soyabean. We use the deseasonalised price for these crops for the further analysis. To determine if there are structural breaks, we conducted the

¹⁵ Refer appendix B for the details on the variance of returns in the portfolio.

Elliot and Muller (2006) test. We did not identify any structural break in spot and futures returns for all thirteen commodities except gram¹⁶ spot price.

We then estimate the conditional mean and conditional variances of the spot and futures returns of the thirteen agricultural commodities. We estimate mean equation by VAR model. We find that the futures returns (with lags) affect current period spot returns for almost all crops. However, that spot returns (with lags) affect current period futures returns for gram, maize, pepper and castorseed only (table 1).

Insert table 1

We now explain the variance model. The table 2 presents the estimates of BEKK model for spot and futures returns¹⁷ for all thirteen agricultural commodities.

Insert table 2

There is a strong ARCH and GARCH effects in both spot and futures markets for almost all crops. There is unidirectional short term volatility spill over from futures to spot market in case of castorseed, coriander, cotton, gram, mustard, pepper, sugar and wheat. Similarly, there is unidirectional short term volatility spill over from spot to futures market in case of chilli, maize, zeera and soyabean. We also find that there is an asymmetry in the patterns of volatility spillover. Considering soyabean markets first, we find that α_{12} and α_{21} are significant at a very high level of significance. The magnitude of α_{21} is less than α_{12} , implying that the impact of shock in the futures market on the soyabean spot market volatility is less than the impact of shock in the spot market on the soyabean futures market. The coefficients β_{12} and β_{21} measure the impact of previous uncertainties. In case of mustard, β_{21} is significantly high whereas β_{12} coefficient is not significant. This implies that the volatility spillover is unidirectional from futures to spot market. β_{22} coefficient is significant, implying that there is GARCH in the futures market for mustard.

¹⁶ We find the presence of structural break in gram spot price. However, we could not find any sudden discontinuous change in gram spot price, and there is no economic justification for structural break during the sample period. We, therefore, did not accommodate the structural break in our model for gram spot price.

¹⁷ First difference of the log of prices are returns.

In case of castorseed, α_{21} is significant whereas α_{12} is insignificant. This indicates that the shock in the castorseed futures market would influence the future uncertainty in the castorseed spot market, while any shock in the castorseed spot market would not affect the future uncertainty in the castorseed futures market. This implies that there is a short term volatility spill over from futures to spot market in case of castorseed. We also find that α_{21} is significant whereas α_{12} is insignificant for coriander, gram, and cotton.

There is unidirectional long term volatility spill over from spot to futures market in case of castorseed, chilli, gram and soyabean. There is unidirectional long term volatility spill over from futures to spot market in case of coriander, mustard and pepper. In sum, there is short term volatility spill over from futures to spot market in case of castorseed, coriander, cotton, gram, mustard, pepper, sugar and wheat. However, in case of chilli, maize, zeera and soyabean, there is short term volatility spill over from spot to futures market. There is long term volatility spill over from spot to futures market in case of castorseed, chilli, gram and soyabean. However, in case of coriander, mustard and pepper, there is long term volatility spill over from futures to spot market.

The finding of our study differs from the findings of Sehgal, Ahmad and Deisting (2014). The authors have analysed the agricultural commodities traded on various exchanges in India. They found that there is there is a short term volatility spill over from spot to futures markets, however, the long term volatility spillover is from futures to spot market. Sehgal, Ahmad and Deisting (2014) study have not removed the predictable component of price process. There are two components of the price formation process: predictable and unpredictable components. We need to remove the predictable component of the price process to measure the volatility, as only unpredictable component measures the volatility (Moeldina et al., 2003). Nonetheless, our finding is consistent with the findings of Kumar and Shollapur (2015). Kumar and Shollapur (2015) have analysed the volatility spillover between spot and futures returns for four agricultural commodities viz., soya oil, soyabean, mustardseed and gram, and found that there is volatility spill over from futures to spot market in case of soya oil, soyabean and gram. However, in case of mustard seed, there is long term volatility spill over from futures to spot market.

In some commodities, past shocks in spot market is affecting the present volatility of futures markets, however the long term volatility spill over is from futures to spot market. This is because the shocks¹⁸ are of different types. Shocks (news) can be RBI monetary policy, government announcement of minimum support price, climatic factor such as monsoon failure, political crisis and international factors such as euro zone crisis, regional crisis. Some shocks have short¹⁹ term persistence, while some shocks have long²⁰ term persistence. In other words, short term volatility spillover may be from spot to futures, however, the long term volatility spillover is from futures to spot market. We will illustrate this point by taking the example. If there is a some minor changes in RBI monetary policy, it will have its impact on the short term volatility spillover. This type of shock will affect the volatility for a short period of time.it will not affect the long run volatility spill over from one market to other. If spot market reacts first to this news of RBI, then volatility spillover will be from spot to futures. However, if futures market reacts first to this news, then volatility spillover is from futures to spot market. This impact of shock will remain for short period.

If there is a major policy change by RBI or government, it will affect the long term volatility spillover from one market to other. Depending on the nature of shocks, the impact of shock will be for short term or long term. We, therefore, find that short term volatility spillover are different from long run volatility spillover.

We also find that the volatility spill over (short run or long run) from one market to other are different for different crops. This is because shocks affect the different crops in different ways. One shock/ news will not have the same impact on all crops. We will illustrate this point by taking the example. If government announces minimum support price for wheat, then it will affect the volatility of wheat and not of barley. Similarly, if there is a change in exchange rate, it will affect the commodities that are exported or imported. These commodities will have greater impact of shocks. While

¹⁸ Shocks are also known as news.

¹⁹ Some shocks have the impact for few hours or one day, and then market comes to its mean price. It means short run persistence.

²⁰ If the shocks have impact for several days, then it will have long run persistence.

other commodities will also be affected due to import of fertilisers. But the impact on other commodities will be less.

We find the evidence of the asymmetric impact of shocks. The coefficients δ_{ii} and δ_{jj} denote the asymmetric impact of own past shocks on conditional variance. δ_{ij} and δ_{ji} measure the cross market asymmetric response. For almost all crops, the sign of own past shocks is affecting the conditional variance of futures returns. However, the sign of own past shocks is not affecting the conditional variance of spot returns of barley, castorseed, chilli, coriander, cotton seed oilcake, gram, sugar, zeera and wheat. In addition, we find the cross market asymmetric response is between gram spot market and gram futures market. Volatility in the gram spot market following a negative shock in the gram futures market is larger than that following a positive shock in the gram futures market, and vice versa. In other crops, there is no cross market asymmetric response between spot and gram futures markets.

We performed the multivariate diagnostic tests for all crops. The Ljung-Box Q-statistic (multivariate generalizations) for standardised residual for barley is 177.107, with the p-value .868. We, therefore, are not able to reject the null hypothesis of no serial correlation. This implies that the VAR-BEKK model is adequate for barley crop. The multivariate residual diagnostic tests for standardised residuals are reported in table 3.

Insert table 3

We find that the VAR-BEKK model is adequate for all crops (table 3). We also did the multivariate ARCH test to check for residual cross-variable ARCH. The test statistic for barley is 47.76 with p value 0.361. This implies that there is multivariate arch effect in case of barley crop. We find that the multivariate arch effect is present in all thirteen crops (table 3). The Ljung-Box Q-statistic (multivariate generalizations) for standardised residuals and multivariate arch effect suggest that the VAR-BEKK model is adequate.

The movement of futures agricultural commodity prices are interdependent because the agricultural commodity prices are linked through substitutability and complementarity in demand and supply. Furthermore, specific macroeconomic factors such as aggregate demand, inflation, exchange rates and interest rate, and

herd behaviour in financial markets causes the co movement of agricultural commodity prices (Malliaris and Urrutia 1996).

5.1. Intercrop volatility spillover analysis

We look into the intercrop analysis by categorising the agricultural crops into two broad categories viz., rabi crops and kharif. In rabi crops we have taken rabi crops, namely barley, wheat, coriander, gram and mustard. Kharif crop includes castorseed, maize and soyabean. We first explain the rabi crops. We estimate the conditional mean and conditional variances of futures returns in rabi crops futures markets. We denote barley, wheat, coriander, gram and mustard as 1, 2, 3, 4 and 5 respectively. We estimate mean equation by VAR model. Table 4 presents the estimates of BEKK model for futures returns in cereals futures markets.

Insert table 4

There is a short term as well as long term volatility spill over from one market to another in case of rabi crops. α_{12} α_{21} α_{13} α_{23} ... measure the short term volatility spillover and β_{12}, β_{21} β_{13}, β_{23} ... measure the long term volatility spillover.

In rabi crops category, α_{14} is significant. This indicates that the shock in the barley futures market would influence the future uncertainty in the gram futures market. The coefficients α_{32} and α_{31} are significant, which mean that the shock in the coriander futures market would influence the future uncertainty in the wheat futures market, and the shock in the coriander futures market would affect the future uncertainty in the barley futures market. We find that $\beta_{41}, \beta_{21}, \beta_{42}, \beta_{13}, \beta_{23}, \beta_{52}, \beta_{34}, \beta_{15}$ and β_{35} , are significant at a high level of significance.

The study also estimate the conditional mean and conditional variances of futures returns in kharif futures markets. We have taken three kharif crops, namely castorseed, maize and soyabean, and have denoted them as 1, 2 and 3 respectively. We estimate mean equation by VAR model. Table 5 presents the estimates of BEKK model for futures returns in spices futures markets.

Insert table 5

We find that the past shocks in castorseed futures market is affecting the present volatility of that market. However, the past shocks in maize and soyabean futures markets is not affecting the present volatility of that market. There is a short term as well as long term volatility spill over from one market to another in case of kharif crops.

In kharif category, the coefficients α_{21} and α_{12} are significant, which mean that the shock in the maize futures market would influence the future uncertainty in the castorseed futures market, and the shock in the castorseed futures market would affect the future uncertainty in the maize futures market. However, the magnitude of α_{12} is greater than α_{21} which implies that the impact of shock in the castorseed futures market on the maize futures market volatility is more than the impact of shock in the maize futures market on the castorseed futures market. Thus, the basic finding of our study is consistent with previous findings of Zhao and Goodwin (2011) and Gardebroek, Hernandez and Robles (2016). Zhao and Goodwin (2011) examined the volatility spill over between corn and soyabean using data from option market in USA. They found the volatility spillover from the corn market to soyabean market, but not from soyabean to corn markets. Gardebroek, Hernandez and Robles (2016) analysed the volatility spill over across the agricultural markets in USA. The authors have analysed the volatility spill over on a daily, weekly and monthly basis, and found volatility spill over across agricultural commodities especially on the weekly and monthly data. The above mentioned studies suggest that volatility spillovers exist across the agricultural commodity markets because the agricultural commodity prices are linked through substitutability and complementarity in demand and supply, share common market information (Malliaris and Urrutia 1996).

5.2 Hedging strategy

Now, we are now going to explain the second objective which is to examine the hedging effectiveness of agricultural commodity futures market. We now put together the results from the multivariate VAR BEKK specification, and comment on certain

portfolio features²¹ such as portfolio weights, optimal hedge ratio and hedge effectiveness index. We report the average value of portfolio weights ($pw_{12,t}$) for spot and futures returns for all thirteen agricultural commodities in table 6.

Insert table 6

In case of barley crop, the average value of $pw_{12,t}$ in a portfolio comprising of barley spot and futures is 0.78 in favour of barley spot. This implies that optimum holding of barley spot in barley spot/futures portfolio is four-fifth, and one-fifth is barley futures. The optimal portfolio weights indicate that investors can minimise risk without reducing expected returns by possessing more barley physical than barley futures contract in their portfolio.

We present the optimal hedge ratios (β_t) values for spot and futures returns for all thirteen agricultural commodities in table 6. Optimal hedge ratio suggest that what proportion of physical Commodity, we should keep in futures contract with opposite position The optimal hedge ratio²² for castor seed is 0.33 (castorseed spot/castorseed futures), implying if a farmer owns 100 tonnes of castorseed physical commodity, then he needs to sell 33 tonnes of castor futures contract to have the same return but less risk.

The average hedge ratio is less than 0.5 for all crops. We present the values of the hedging effectiveness index for spot and futures returns for all thirteen agricultural commodities in table 6. HEI index suggest the reduction in the risk of hedged portfolio when we compare it with unhedged portfolio. In case of barley, the HEI is 21.90, implying if we hedge barley crop, then we reduce risk by more than 20 percent. There exist volatility spillover between spot and futures agricultural commodity markets (volatility paper), and there is co-movement of agricultural futures and spot prices due to arbitrageurs actions. These two factors facilitate the markets participants to hedge by taking opposite position in futures market (Chang et al. 2011). Thus, the basic

²¹ Optimal portfolio weights is from the perspective of investors, mill owners, cooperative societies such as nafed, etc. Optimal hedge ratio is from the perspective of farmers whereas hedge effectiveness index is for all market participants.

²² OHR implies one rupee long in castorseed spot should be shorted by 0.33 paisa in castorseed futures.

finding of our study is consistent with previous findings of Choudhary (2009), Mann and Sephton (2010) and Aggrawal, Jain and Thomas (2014).

We also examined (ref section 5.1) the interdependence of futures prices of various crops traded on the national commodity exchanges. A finding of significant linkages between the agricultural commodities, would imply the existence of cross hedging opportunities. We now present the results of cross hedging in the next sub section, i.e. intercrop analysis.

5.2.1. Intercrop hedging analysis

We report the average value of $pw_{12,t}$ based on model VAR BEKK, for futures returns for rabi and kharif crops in table 7 and 8 respectively.

Insert table 7 and 8

In case of gram/barley futures returns, the average value of $pw_{12,t}$ in a portfolio comprising of gram futures and barley futures returns is 0.50 in favour of gram futures. This implies that optimum holding of gram futures in gram/barley futures portfolio is approximately half and half for barley futures.

We present the optimal hedge ratios (β_t) values for futures returns for rabi and kharif crops in table 7 and 8. The optimal hedge ratio for castorseed/soyabean futures is 0.19 (castorseed futures/soyabean futures), it tell us that if we buy 100 tonnes of castorseed futures contract then we need to sell 19 tonnes if soyabean contract to have the same return but less risk.

We present the values of the hedging effectiveness index for futures returns for rabi and kharif crops in tables 7 and 8. HEI index suggest the reduction in the risk of hedged portfolio when we compare it with unhedged portfolio. The HEI for castorseed/soyabean is 28.40. The means that when we hedge castorseed against soyabean, then risks in holding castorseed get reduced by 28.40 percent. Thus, the basic finding of our study is consistent with previous findings of Lahiani, Nguyen and Vo (2013). Lahiani, Nguyen and Vo (2013) examined the volatility spillover across the four agricultural commodities viz., sugar, wheat, corn and cotton. The study found the

existence volatility spillovers across the agricultural commodity markets, and computed optimal hedge ratios and optimal portfolio weights to minimise the risk of agricultural commodity portfolio. There exist cross hedging across crops because the agricultural commodity prices are linked through substitutability and complementarity in demand and supply, share common market information (Malliaris and Urrutia 1996).

5.3. Robustness Analysis

Next, we perform entire estimations again with weekly data to further evaluate the robustness of our results. These include re-examining our BEKK model and hedging effectiveness of futures markets in and across the agricultural commodities over the weekly data. We estimate the conditional mean and conditional variances of the spot and futures returns of the thirteen agricultural commodities. We estimate mean equation by VAR model. We find that the futures returns (with lags) affect current period spot returns for castorseed, maize, mustard, soyabean and sugar. However, that spot returns (with lags) affect current period futures returns for barley only (table 9).

The table 10 presents the estimates of BEKK model for spot and futures returns for all thirteen agricultural commodities over weekly data.

Insert table 9 and 10

There is a strong ARCH and GARCH effects in both spot and futures markets for almost all crops. Thus, ARCH and GARCH results based on weekly data are similar to results based on daily data. There is unidirectional short term volatility spill over from futures to spot market in case of castorseed, chilli, coriander, cotton seed oilcake, gram, pepper and sugar. Similarly, there is unidirectional short term volatility spill over from spot to futures market in case of chilli, coriander, gram, maize, mustard, zeera and wheat. We also find that there is an asymmetry in the patterns of volatility spillover. Considering mustard markets first, we find that α_{12} and α_{21} are significant at a very high level of significance. The magnitude of α_{21} is less than α_{12} , implying that the impact of shock in the futures market on the mustard spot market volatility is less than the impact of shock in the spot market on the mustard futures market.

While the daily as well as weekly results suggest the volatility spill over between the spot and futures commodities, the volatility spillover in some of the crops varies for weekly and daily data. All the elements for parameter matrices A and B for all crops, are not significant even at ten percent level of significance. Thus, while the results using daily data indicate that all the elements for parameter matrices A and B for sugar crop are significant, we do not observe this with weekly data.

There is unidirectional long term volatility spill over from spot to futures market in case of barley, coriander, chilli, gram, maize, sugar wheat and zeera. There is unidirectional long term volatility spill over from futures to spot market in case of castorseed, cotton seed oilcake, mustard and pepper.

At the weekly level, we have also re-examined the second objective which is to examine the hedging effectiveness of agricultural commodity futures market. We report the average value of portfolio weights ($pw_{12,t}$) for spot and futures returns for all thirteen agricultural commodities in table 11.

Insert table 11

In case of barley crop, the average value of $pw_{12,t}$ in a portfolio comprising of barley spot and futures is 0.946 in favour of barley spot. This implies that optimum holding of barley spot in barley spot/futures portfolio is almost 100 percent. The optimal portfolio weights indicate that investors can minimise risk without reducing expected returns by possessing more physical commodities than futures contract in their portfolio. This is similar to our results based on daily data for all crops.

We present the optimal hedge ratios (β_t) values for spot and futures returns for all thirteen agricultural commodities in table 11. The optimal hedge ratio for gram is 0.35 (gram spot/ gram futures), implying if a farmer owns 100 tonnes of gram physical commodity, then he needs to sell 35 tonnes of gram futures contract to have the same return but less risk. For all crops, optimal hedge ratio values based on weekly data are almost similar to the results based on daily data,

In case of barley, the HEI is 23.52, implying if we hedge barley crop, then we reduce risk by more than 23 percent. For pepper crop, we can reduce risk by approximately 52 percent where as in case of mustard we can reduce risk by 55 percent. We find that the weekly HEI index is greater than the daily HEI index for almost all crops (table

11). This implies that weekly HEI index suggest greater reduction in the risk of hedged portfolio when we compare it with unhedged portfolio.

Overall, the robustness test results show that there are significant interrelations in conditional volatility in and across agricultural commodities analysed on a weekly and daily basis. We conclude that the hedging effectiveness of agricultural commodity futures market improves with the weekly data.

6. Conclusions and policy implications

Using daily data on spot and futures returns for wheat, barley, maize, gram, mustard, castorseed, soyabean, zeera, chilli, coriander, pepper, cotton seed oilcake and sugar, we simultaneously model the shocks in spot and futures returns. We estimate the volatility spillover effects for each of the futures and spot agricultural commodity prices, in and across all markets. We employ VAR-BEKK specification to model the volatility spillover. Based on the results from VAR-BEKK model, we calculate optimal portfolio weights, optimal hedge ratios, and hedge effectiveness index to examine the hedging effectiveness of futures markets.

The findings of the study are as follows:

First, we find that the past shocks in one market is affecting the present volatility of that market, and this is true for almost all markets. In case of almost all crops, the past volatility in one market is affecting the present volatility of that market. There is short term volatility spill over from futures to spot market in case of castorseed, coriander, cotton, gram, mustard, pepper, sugar and wheat. However, in case of chilli, maize, zeera and soyabean, there is short term volatility spill over from spot to futures market.

There is long term volatility spill over from futures to spot market in case of coriander, mustard and pepper. However, in case of castorseed, chilli, gram and soyabean, there is long term volatility spill over from spot to futures market. This is because the shocks (news) are of different types. While impact of some shocks persists for few hours, for others, it persists for several days. If spot market reacts first to the news,

then volatility spillover will be from spot to futures. However, if futures market reacts first to news, then volatility spillover is from futures to spot market.

We also find that the volatility spill over (short run or long run) from one market to other are different for different crops. This is because shocks affect different crops in different ways. One shock/news will not have the same impact on all crops. We will illustrate this point by taking the example. If government announces minimum support price for wheat, then it will affect the volatility of wheat and not of barley.

Second, we find evidence of asymmetric impact of shocks. This means that bad news and good news (of equal magnitude) have different impact on the volatility of returns. We also find the cross market asymmetric response is between gram spot market and gram futures market. Volatility in the gram spot market following a negative shock in the gram futures market is larger than that following a positive shock in the gram futures market, and vice versa. In other crops, there is no cross market asymmetric response between spot and gram futures markets.

Third, the agricultural commodity prices are linked through substitutability and complementarity in demand and supply, share common market information, and are affected by herd behaviour and specific macroeconomic factors such as aggregate demand, inflation, interest rate, etc. (Malliaris and Urrutia 1996). We, therefore, analyse the intercrop analysis by categorising the agricultural crops into two broad categories, namely rabi crops and kharif crops. In rabi crops we have taken five crops - barley, wheat, coriander, gram and mustard, and kharif crop includes castorseed, maize and soyabean. We find that the past shocks in wheat futures market is affecting the present volatility of that market, and this is true for coriander as well. There is a short term as well as long term volatility spill over from one market to another in case of rabi crops and kharif crops as well.

Fourth, we estimate the conditional covariances from multivariate VAR-BEKK model to compute the optimal portfolio weights and time varying optimal hedge ratios, which would indicate hedging effectiveness of agricultural commodities futures markets. In addition, we calculated the hedging effectiveness index. A high value of HEI implies high hedging effectiveness and greater reduction in risk. The evidence provided by optimal portfolio weights calculated from the multivariate model for almost all crops

suggest that we should hold physical commodity in greater proportion than futures contracts. This implies that investors can minimise risk without reducing expected returns by possessing more physical commodities than futures contract in their portfolio. The time varying OHRs calculated from the multivariate model are less than 0.5²³ for almost all crops. The HEI for barley suggest that if we hedge barley crop, then we reduce risk by more than 20 percent. To sum up, we find that the hedging effectiveness of futures market is high in agricultural commodities, however, there is a wide variation across the crops.

Lastly, we also perform entire estimations again with weekly data to further evaluate the robustness of our results. Overall, the robustness test results show that there are significant interrelations in conditional volatility in and across agricultural commodities analysed on a weekly basis. We find that the weekly HEI index is greater than the daily HEI index for almost all crops. This implies that weekly HEI index suggest greater reduction in the risk of hedged portfolio when we compare it with unhedged portfolio. We conclude that the hedging effectiveness of agricultural commodity futures market improves with the weekly data.

This paper adds to the existing literature in the following ways: Firstly, this paper adds to the optimum hedge ratio literature by estimating the hedge ratio using multivariate GARCH approach. Since there are significant linkages²⁴ between agricultural commodity markets, so we cannot use univariate GARCH model which assume independence, and fails to model the interdependencies between the prices. Secondly, previous studies used multivariate Generalised autoregressive conditional heteroskedastic (GARCH) model to estimate time varying hedge ratios for agricultural commodity markets (Choudhary 2009; Bekarman 2011). However, past studies did not examine the volatility spillover effects for each of the futures and spot agricultural commodity prices in and across all markets. Thirdly, this study has examined not only the hedging between spot and futures prices of a particular crop, but across the futures prices of rabi and kharif crops. Past studies have not done so. Lastly, this

²³ Optimal hedge ratio tell us what proportion of physical Commodity, we should keep in futures contract with opposite position

²⁴ The agricultural commodity markets are linked through substitutability and complementarity.

study has calculated the hedge effectiveness index. Past studies did not consider hedge effectiveness index for risk reduction.

The following policy implications result from this paper: firstly, volatility spill over effects between markets keeps on changing. Volatility spill over effects are harmful²⁵ for economic performances. The authorities need to start awareness programme among the farmers (and other market participants) about the benefits of futures market so that farmers (and other market participants) understand the origins²⁶ and drivers of volatility across markets and, accordingly manage risk in agricultural commodity markets. Secondly, Authorities need to be careful in making policy changes in one commodity market because price volatility in one market spillover to the price volatility of other markets through market channels. In addition, Optimum hedging strategies improve the hedging role of futures markets and add to social welfare. Agricultural producers (market participants) improve their utility by reducing their portfolio variance (Bekarman 2011).The reduction in portfolio variance is equivalent to improvement in utility for the market participants. We can use futures market, and compute optimal portfolio weights and hedge effectiveness index to maximise social welfare. This is the social implication of this paper.

²⁵ It is harmful because investors equate higher volatility with greater risk, which in turn affects their investment decisions.

²⁶ If the volatility is from futures (spot) to spot (futures), we can follow the optimum hedging strategy to protect ourselves from excessive volatility.

Tables

Table 1. Mean model estimates of daily spot and futures returns in agricultural commodity markets, 2009-2014

Variable	Barley	Castor Seed	Chilli	Coriander	Cotton	Gram	Maize	Mustard	Pepper	Soya bean	Sugar	Wheat	Zeera
Mean Model (Spot Price)													
SP{1}	0.163*** (0.029)	-0.072*** (0.025)	0.114*** (0.032)	0.030 (0.026)	-0.007 (0.024)	-0.083*** (0.027)	0.054* (0.028)	-0.055** (0.026)	-0.184*** (0.033)	-0.031 (0.022)	0.030 (0.029)	0.127*** (0.024)	-0.052* (0.027)
SP{2}	0.023 (0.026)		0.072*** (0.026)	0.003 (0.023)	0.060*** (0.022)	-0.038 (0.024)	0.057** (0.027)	-0.040* (0.021)	0.032 (0.026)		0.021 (0.028)		0.026 (0.026)
SP{3}	-0.004 (0.025)		0.032 (0.024)		0.052** (0.022)								0.053** (0.026)
SP{4}			0.003 (0.022)		0.047** (0.021)								0.048** (0.024)
SP{5}					0.076*** (0.020)								
SP{6}					0.018 (0.020)								
FP{1}	0.094*** (0.014)	0.239 (0.019)	0.093*** (0.011)	0.147*** (0.016)	0.226*** (0.014)	0.433*** (0.025)	0.036** (0.014)	0.341*** (0.020)	0.314*** (0.017)	0.276*** (0.019)	0.131*** (0.016)	0.029** (0.012)	0.014*** (0.011)
FP{2}	0.025* (0.013)		0.022** (0.010)	0.092*** (0.016)	-0.007 (0.015)	-0.037 (0.025)	0.049*** (0.014)	0.113*** (0.020)	0.049*** (0.017)		0.090*** (0.016)		0.138 (0.011)
FP{3}	0.018 (0.014)		0.005 (0.010)		-0.022 (0.013)								-0.008 (0.011)
FP{4}			0.002 (0.010)		-0.022 (0.014)								-0.20 (0.011)
FP{5}					-0.043*** (0.014)								
FP{6}					-0.007 (0.013)								
Const	0.039*** (0.014)	0.030 (0.021)	-0.045** (0.019)	0.031 (0.032)	0.029* (0.018)	0.048** (0.022)	0.019 (0.015)	0.028* (0.015)	0.027 (0.017)	0.046** (0.021)	-0.017* (0.010)	0 (0.009)	-0.010 (0.011)
Mean Model (Futures Price)													
SP{1}	0.046 (0.044)	0.053* (0.032)	0.149** (0.064)	0.032 (0.034)	-0.065* (0.038)	0.030 (0.028)	0.043 (0.039)	0.010 (0.025)	0.057 (0.065)	0.078*** (0.028)	0.035 (0.038)	0.047* (0.027)	-0.028 (0.059)

Table 1 (Continued)

Variable	Barley	Castor seed	Chilli	Cori ander	Cotton	Gram	Maize	Mustard	Pepper	Soya bean	Sugar	Wheat	Zeera
SP{2}	-0.043 (0.043)		-0.072 (0.060)	-0.062* (0.033)	-0.052 (0.036)	0.012 (0.028)	0.065 (0.041)	0.013 (0.022)	0.031 (0.050)		0.032 (0.037)		0.126* (0.066)
SP{3}	-0.053 (0.046)		-0.032 (0.058)		-0.042 (0.036)								0.045 (0.069)
SP{4}			-0.003 (0.060)		0.009 (0.035)								0.002 (0.052)
SP{5}					0.013 (0.033)								
SP{6}					-0.003 (0.033)								
FP{1}	-0.027 (0.026)	-0.013 (0.026)	-0.061** (0.029)	0.033 (0.026)	0.033 (0.025)	0.056** (0.026)	0.022 (0.026)	-0.023 (0.024)	-0.069** (0.031)	-0.027 (0.024)	0.004 (0.028)	-0.035* (0.019)	0.029 (0.027)
FP{2}	-0.004 (0.024)		0.018 (0.027)	-0.025 (0.028)	0.075*** (0.026)	0.017 (0.029)	-0.042* (0.025)	0.060*** (0.023)	-0.071** (0.034)		-0.032 (0.028)		0.007 (0.029)
FP{3}	-0.002 (0.025)		-0.014 (0.026)		0.070*** (0.023)								0.036 (0.028)
FP{4}			-0.058 (0.027)		0.035 (0.025)								-0.039 (0.027)
FP{5}					0.009 (0.023)								
FP{6}					0.036 (0.023)								
Const	-0.005 (0.027)	0.015 (0.032)	-0.050 (0.049)	-0.043 (0.047)	-0.065** (0.030)	0.030 (0.026)	-0.008 (0.026)	0.027 (0.020)	0.074** (0.036)	0.072** (0.030)	-0.046** (0.018)	0 (0.015)	-0.046 (0.028)

Source: Author's estimation based on secondary data from the National Commodity and Derivative Exchange (NCDEX), Mumbai, and the Multi-Commodity Exchange (MCX), Mumbai.

Notes: All variables are the first difference of the log of prices . First difference of the log of prices are returns. SP denotes spot returns and FP denotes futures returns. "Cotton" refers to cotton seed oil cake. Period of analysis: May 2009 – August 2014 for all crops (daily data). Optimal lags are taken for all crops. {*i*} denotes the lag *i* where *i* = 1,2,3, 4 ...

We have deflated the data with wholesale price index. We have used the seasonal dummies to remove the impact of seasonality from the spot and futures prices.

***, **, and * indicate statistical significance at the 1%, 5% and 10% levels, respectively. Values in small brackets denote standard errors, Mean equation is VAR model. Estimations based on VAR-BEKK model.

Table 2. Variance model estimates of daily spot and futures returns in agricultural commodity markets, 2009-2014

Variable	Barley	Castor seed	Chilli	Cori Ander	Cotton	Gram	Maize	Mustard	Pepper	Soya bean	Sugar	Wheat	Zeera
c_{11}	0.109*** (0.036)	0.024 (0.108)	0 (0.619)	0.601*** (0.054)	0.127*** (0.032)	0.546*** (0.061)	0.037 (0.041)	0 (0.253)	0.066 (0.138)	0.285*** (0.031)	0.045*** (0.017)	0.165*** (0.025)	0 (0.042)
c_{21}	-0.006 (0.048)	-0.084 (0.133)	0.613*** (0.123)	-0.129 (0.092)	0.069 (0.070)	0.593*** (0.059)	0.014 (0.021)	0.413*** (0.053)	-0.269*** (0.067)	0.274*** (0.070)	0.012 (0.025)	0.012 (0.042)	-0.143** (0.062)
c_{22}	0.502*** (0.142)	0 (0.0520)	0.502*** (0.193)	0 (1.066)	0.151*** (0.057)	0 (0.114)	0.099* (0.054)	0.101*** (0.037)	0.309*** (0.114)	0 (0.138)	0.073*** (0.025)	0.019 (0.16)	0.051*** (0.145)
α_{11}	0.291*** (0.085)	0.190*** (0.034)	0.561*** (0.134)	0.263*** (0.039)	0.271*** (0.044)	0.276*** (0.052)	0.162*** (0.035)	0.468*** (0.053)	-0.178** (0.073)	0.099** (0.041)	0.289*** (0.038)	0.428*** (0.048)	-0.091* (0.053)
α_{12}	-0.130 (0.287)	0.041 (0.047)	0.693*** (0.195)	0.062 (0.043)	-0.014 (0.048)	0.004 (0.057)	0.064* (0.036)	0.045** (0.020)	0.195* (0.113)	0.191*** (0.072)	0.222*** (0.050)	0.159*** (0.054)	0.411*** (0.095)
α_{21}	0.006 (0.029)	-0.100*** (0.031)	0.020 (0.062)	0.081** (0.036)	-0.039* (0.020)	-0.263*** (0.073)	-0.003 (0.021)	-0.150** (0.068)	0.256*** (0.028)	0.164*** (0.028)	0.062** (0.030)	-0.018 (0.029)	0.078*** (0.018)
α_{22}	0.061 (0.102)	0.149*** (0.041)	-0.120 (0.124)	0.357*** (0.053)	0.148*** (0.037)	-0.103 (0.067)	-0.222*** (0.031)	-0.111*** (0.035)	0 (0.065)	-0.081 (0.064)	-0.198*** (0.037)	-0.155*** (0.022)	-0.034 (0.058)
β_{11}	0.094*** (0.044)	0.751*** (0.017)	0.524*** (0.142)	-0.696*** (0.014)	0.948*** (0.016)	0.798*** (0.055)	0.981*** (0.010)	0.694*** (0.067)	0.668*** (0.130)	0.966*** (0.030)	0.933*** (0.014)	0.879*** (0.026)	0.880*** (0.093)
β_{12}	0.232 (0.169)	-0.478*** (0.028)	-0.697*** (0.203)	-1.244*** (0.065)	0.017 (0.015)	-0.137*** (0.044)	0.001 (0.009)	-0.002 (0.007)	0.352 (0.230)	0.211*** (0.043)	-0.048** (0.22)	-0.038 (0.026)	0.374 (0.240)
β_{21}	-0.019 (0.043)	0.223*** (0.011)	0.048*** (0.018)	-0.205*** (0.024)	0.001 (0.007)	-0.123*** (0.018)	0 (0.005)	0.062*** (0.022)	0.105* (0.055)	-0.067*** (0.022)	0.015* (0.008)	0.017 (0.014)	0.043 (0.038)
β_{22}	0.778*** (0.136)	1.106*** (0.014)	1.027*** (0.021)	0.744*** (0.016)	0.971*** (0.008)	0.872*** (0.021)	0.971*** (0.009)	0.989*** (0.003)	0.855*** (0.089)	0.854*** (0.025)	0.968*** (0.009)	0.993*** (0.006)	0.827*** (0.106)
δ_{11}	-0.144 (0.135)	0.032 (0.048)	0.162 (0.142)	0.014 (0.124)	-0.115 (0.077)	-0.141 (0.088)	0.127* (0.069)	-0.268** (0.125)	-0.294*** (0.072)	0.178*** (0.063)	0.069 (0.134)	0.034 (0.135)	0.206 (0.073)
δ_{12}	-0.276 (0.170)	0.080 (0.072)	0.240 (0.249)	0.152 (0.126)	-0.168*** (0.064)	-0.192** (0.093)	-0.107* (0.062)	0.030 (0.041)	0.024 (0.0175)	-0.305*** (0.079)	-0.106 (0.128)	-0.123 (0.082)	0.024 (0.166)

Table 2 (Continued)

Variable	Barley	Castor seed	Chilli	Cori Ander	Cotton	Gram	Maize	Mustard	Pepper	Soya bean	Sugar	Wheat	Zeera
δ_{21}	0.011 (0.034)	-0.241*** (0.036)	0.080 (0.070)	0.120** (0.060)	0.034 (0.032)	0.218** (0.087)	0.023 (0.034)	0.277*** (0.078)	0.104 (0.052)	-0.049 (0.054)	-0.106** (0.043)	-0.094 (0.047)	-0.064*** (0.025)
δ_{22}	-0.335*** (0.101)	-0.132** (0.056)	-0.015 (0.157)	-0.286*** (0.093)	0.235*** (0.051)	0.251*** (0.088)	0.129 (0.089)	-0.116** (0.047)	0.107 (0.108)	0.294*** (0.063)	-0.175** (0.074)	0.025 (0.061)	0.182*** (0.068)

Source: Author's estimation based on secondary data from the National Commodity and Derivative Exchange (NCDEX), Mumbai, and the Multi-Commodity Exchange (MCX), Mumbai.

Notes: All variables are the first difference of the log of prices . First difference of the log of prices are returns. SP denotes spot returns and FP denotes futures returns. "Cotton" refers to cotton seed oil cake. Period of analysis: May 2009 – August 2014 for all crops (daily data). Optimal lags are taken for all crops. Estimations based on VAR-BEKK model. We denotes spot returns as 1 and futures returns as 2.

We have deflated the data with wholesale price index. We have used the seasonal dummies to remove the impact of seasonality from the spot and futures prices.

***, **, and * indicate statistical significance at the 1%, 5% and 10% levels, respectively. Values in small brackets denote standard errors, and we have not reported constants in the table to save space.

$C = \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix}$; $A = \begin{bmatrix} \alpha_{11} & \alpha_{12} \\ \alpha_{21} & \alpha_{22} \end{bmatrix}$ $B = \begin{bmatrix} \beta_{11} & \beta_{12} \\ \beta_{21} & \beta_{22} \end{bmatrix}$; $D = \begin{bmatrix} \delta_{11} & \delta_{12} \\ \delta_{12} & \delta_{22} \end{bmatrix}$ denotes the coefficient matrices estimated from equation(6).

The coefficients α_{11} and α_{22} measure the impact of past shocks on the present volatility of the same time series variable.

α_{12} represents the impact of past shocks of first variable (spot) on the present volatility of the second (futures) variable. β_{12} and β_{21} measure the impact of previous uncertainties. The coefficients δ_{ii} and δ_{jj} denote the asymmetric impact of own past shocks on conditional variance.

δ_{ij} and δ_{ji} measure the cross market asymmetric response.

Table 3. Multivariate Diagnostic tests for standardized residuals (daily), 2009-2014

Variable	Barley	Castor seed	Chilli	Cori ander	Cotton	Gram	Maize	Mustard	Pepper	Soya bean	Sugar	Wheat	Zeera
Multivariate test for model adequacy													
LB-Q(40)	177.107	154.407	185.095	163.569	187.676	156.045	181.555	183.978	158.622	150.605	178.753	164.388	185.143
Prob	0.868	0.609	0.084	0.407	0.066	0.573	0.116	0.094	0.515	0.690	0.147	0.389	0.084
Test for Multivariate ARCH													
statistic	47.76	31.79	46.65	42.24	24.88	36.48	21.62	40.79	506.60	40.68	24.68	21.19	42.75
Prob	0.361	0.931	0.404	0.589	0.581	0.105	0.998	0.650	0.168	0.655	0.99	0.999	0.56

Source: Author's estimation based on secondary data from the National Commodity and Derivative Exchange (NCDEX), Mumbai, and the Multi-Commodity Exchange (MCX), Mumbai.

Notes: All variables are the first difference of the log of prices. First difference of the log of prices are returns. Period of analysis: May 2009 – August 2014 for all crops (daily data). Optimal lags are taken for all crops.

We have deflated the data with wholesale price index. We have used the seasonal dummies to remove the impact of seasonality from the spot and futures prices.

Diagnostic tests based on VAR-BEKK model.

LB-Q(40) denotes for the Ljung- Box Q-statistic (multivariate generalizations) for standardised residuals up to 40 lags.

In LB-(Q) test, the null hypothesis is that the model is adequate or no serial correlation.

Multivariate ARCH test is used to checks for residual cross-variable ARCH.

Table 4. Mean and variance model estimates of daily rabi crops futures returns, 2009-2014

	Barley	Wheat	Coriander	Gram	Mustard
Barley lag1	0.024*** (0.007)	0.040* (0.023)	0.116** (0.057)	-0.033 (0.027)	0.004 (0.035)
Wheat lag1	0.134** (0.053)	0.006 (0.048)	-0.039 (0.646)	0.090* (0.047)	0.017 (0.043)
Coriander lag1	-0.021 (0.019)	-0.004 (0.014)	0 (0.040)	-0.018 (0.018)	-0.014 (0.014)
Gram lag1	0.050 (0.037)	-0.008 (0.020)	0.035 (0.066)	-0.039 (0.033)	0.052* (0.031)
Mustard lag1	0.013 (0.052)	-0.003 (0.025)	0.011 (0.067)	0.068* (0.041)	-0.049 (0.033)
Constant	-0.041 (0.029)	0.021 (0.025)	-0.022 (0.061)	0.006 (0.031)	0.026 (0.029)
Variance Model					
	Barley ($i = 1$)	Wheat ($i = 2$)	Coriander ($i = 3$)	Gram ($i = 4$)	Mustard ($i = 5$)
α_{i1}	0.053 (0.113)	0.147 (0.135)	0.060* (0.032)	-0.147** (0.073)	0.048 (0.152)
α_{i2}	-0.065 (0.063)	0.277*** (0.054)	-0.047** (0.019)	0.092** (0.042)	-0.102** (0.044)
α_{i3}	-0.028 (0.141)	0.037 (0.127)	0.332*** (0.077)	-0.156 (0.139)	0.117 (0.117)
α_{i4}	0.145** (0.073)	0.006 (0.100)	-0.027 (0.038)	-0.010 (0.074)	-0.303*** (0.080)
α_{i5}	-0.002 (0.066)	0.023 (0.052)	0.006 (0.020)	-0.116 (0.080)	0.021 (0.075)
β_{i1}	-0.105 (0.265)	0.276* (0.164)	-0.051 (0.083)	-0.537** (0.240)	-0.101 (0.461)
β_{i2}	0.005 (0.079)	0.924*** (0.044)	0.029 (0.023)	-0.127* (0.070)	0.009 (0.080)
β_{i3}	-0.608** (0.262)	0.292** (0.146)	0.858*** (0.094)	-0.007 (0.427)	-0.639** (0.333)
β_{i4}	-0.130 (0.791)	0.033 (0.135)	0.172** (0.097)	-0.470 (0.687)	0.104 (0.317)

Table 4: variance model (Continued)

	Barley ($i = 1$)	Wheat ($i = 2$)	Coriander ($i = 3$)	Gram ($i = 4$)	Mustard ($i = 5$)
β_{i5}	-0.589** (0.257)	0.161 (0.135)	0.140*** (0.053)	-0.512 (0.327)	0.591** (0.251)
δ_{i1}	0.285 (0.472)	0.167 (0.477)	-0.138** (0.060)	-0.048 (0.149)	0.163 (0.278)
δ_{i2}	-0.043 (0.090)	0.085 (0.160)	0.043 (0.047)	0.139** (0.061)	-0.118 (0.085)
δ_{i3}	-0.139 (0.352)	0.365 (0.371)	-0.181 (0.147)	-0.034 (0.170)	-0.111 (0.273)
δ_{i4}	0.242** (0.096)	-0.206 (0.142)	-0.048 (0.140)	0.120 (0.190)	0.039 (0.161)
δ_{i5}	-0.049 (0.277)	0.033 (0.220)	0.011 (0.094)	0.208 (0.151)	0.172 (0.118)
LB-Q(5)	127.715				
Prob	0.415				
Multivariate ARCH statistic	1174.17				
Prob	0.150				

Source: Author's estimation based on secondary data from the National Commodity and Derivative Exchange (NCDEX), Mumbai, and the Multi-Commodity Exchange (MCX), Mumbai.

Notes: All variables are the first difference of the log of futures prices. First difference of the log of futures prices are futures returns. Period of analysis: May 2009 – August 2014 for all crops(daily data). We denote barley, wheat, coriander, gram,zeera and mustard as 1, 2,3,4,5 and 6 respectively. Optimal lags are taken according to information criteria.

We have deflated the data with wholesale price index. We have used the seasonal dummies to remove the impact of seasonality from the spot and futures prices.

***, **, and * indicate statistical significance at the 1%, 5% and 10% levels, respectively. Mean equation is VAR model. Estimations based on VAR-BEKK model.

The coefficients α_{11} and α_{22} measure the impact of past shocks on the present volatility of the same time series variable. α_{12} represents the impact of past shocks of first variable on the present volatility of the second variable. β_{12} and β_{21} measure the impact of previous uncertainties.

The coefficients δ_{ii} and δ_{jj} denote the asymmetric impact of own past shocks on conditional variance.

δ_{ij} and δ_{ji} measure the cross market asymmetric response.

Table 5. Mean and variance model estimates of daily kharif crops futures returns, 2009-2014

Mean Model			
	Castorseed	Maize	Soyabean
Castorseed lag1	0.060** (0.028)	0.027 (0.017)	-0.006 (0.026)
Castorseed lag2	-0.003 (0.025)	0.010 (0.018)	-0.004 (0.025)
Maize lag1	-0.020 (0.039)	0.039 (0.029)	0.017 (0.034)
Maize lag2	-0.051 (0.037)	-0.006 (0.027)	0.045 (0.035)
Soyabean lag1	-0.034 (0.032)	0.033 (0.020)	0.017 (0.031)
Soyabean lag2	-0.005 (0.032)	0.052 (0.021)	0.055** (0.027)
Constant	-0.002 (0.045)	-0.001 (0.028)	0.061 (0.038)
Variance Model			
	Castorseed ($i = 1$)	Maize ($i = 2$)	Soyabean ($i = 3$)
α_{i1}	0.264*** (0.048)	-0.169** (0.071)	-0.008 (0.043)
α_{i2}	0.084*** (0.030)	0.058 (0.103)	0.068** (0.033)
α_{i3}	0.039 (0.028)	-0.124 (0.079)	0.111 (0.047)
β_{i1}	0.870*** (0.063)	0.440*** (0.153)	-0.081 (0.072)
β_{i2}	0.001 (0.056)	0.690*** (0.035)	-0.662*** (0.027)
β_{i3}	-0.242 (0.091)	1.103*** (0.046)	0.318*** (0.036)
δ_{i1}	0.029 (0.070)	0.047 (0.116)	-0.007 (0.051)
δ_{i2}	-0.003 (0.040)	-0.014 (0.097)	0.017 (0.032)
δ_{i3}	0.027 (0.045)	0.281*** (0.090)	0.010 (0.065)
LB-Q(5)	27.236		
Prob	0.983		
Multivariate ARCH statistic	202.69		
Prob	0.118		

Source: Author's estimation based on secondary data

Notes: All variables are the first difference of the log of futures prices (returns). We denote castorseed, maize, and soyabean as 1, 2 and 3 respectively. ***, **, and * indicate statistical significance at the 1%, 5% and 10% levels, respectively. Mean equation is VAR model.

The coefficients α_{11} and α_{22} measure the impact of past shocks on the present volatility of the same time series variable. α_{12} represents the impact of past shocks of first variable on the present volatility of the second variable. β_{12} and β_{21} measure the impact of previous uncertainties. Estimations based on VAR-BEKK model. The coefficients δ_{ii} and δ_{jj} denote the asymmetric impact of own past shocks on conditional variance.

δ_{ij} and δ_{ji} measure the cross market asymmetric response.

Table 6. Portfolio features for daily spot and futures returns, 2009-2014

Barley	Castor seed	Chilli	Coriander	Cotton	Gram	Maize	Mustard	Pepper	Soya bean	Sugar	Wheat	Zeera
Optimal portfolio weights (average)												
0.780	0.852	0.997	0.779	0.829	0.606	0.763	0.620	1.000	0.954	0.742	0.724	0.996
Average optimal hedge ratio												
0.223	0.331	0.170	0.175	0.224	0.325	0.057	0.327	0.280	0.491	0.128	0.166	0.158
Hedge effectiveness (%)												
21.956	24.544	16.777	7.533	13.125	13.881	0.951	14.528	29.578	46.834	3.839	5.755	15.497

Source: Author's estimation based on secondary data from the National Commodity and Derivative Exchange (NCDEX), Mumbai, and the Multi-Commodity Exchange (MCX), Mumbai.

Notes: All variables are the first difference of the log of prices. First difference of the log of prices are returns. Period of analysis: May 2009 – August 2014 for all crops (daily data) . Optimal lags are taken for all crops.

We have deflated the data with wholesale price index. We have used the seasonal dummies to remove the impact of seasonality from the spot and futures prices.

Mean equation is VAR model. The portfolio weights are reported for spot agricultural commodities. Calculations based on VAR-BEKK model.

Table 7. Portfolio features for daily rabi crops futures returns, 2009-2014

Crops combination	Optimal portfolio weights (average)	Average optimal hedge ratio	Hedge effectiveness (%)
Barley/Wheat	0.303	0.273	3.859
Barley/Coriander	0.816	0.126	4.967
Barley/Gram	0.500	0.213	4.565
Barley/Coriander	0.363	0.341	7.793
Wheat/Coriander	0.876	0.030	0.546
Wheat/Gram	0.677	0.071	1.006
Wheat/Mustard	0.576	0.131	2.252
Coriander/Gram	0.178	0.422	5.794
Coriander/Mustard	0.115	0.454	4.478
Gram/Mustard	0.342	0.460	14.210

Source: Author's estimation based on secondary data from the National Commodity and Derivative Exchange (NCDEX), Mumbai, and the Multi-Commodity Exchange (MCX), Mumbai.

Notes: All variables are the first difference of the log of futures prices. First difference of the log of futures prices are futures returns. "Period of analysis: May 2009 – August 2014 for all crops(daily data). Optimal lags are taken for all crops.

We have deflated the data with wholesale price index. We have used the seasonal dummies to remove the impact of seasonality from the spot and futures prices.

Mean equation is VAR model. Calculations based on VAR-BEKK model.

Table 8. Portfolio features for daily Kharif crops futures returns

	Optimal portfolio weights (average)	Average optimal hedge ratio	Hedge effectiveness (%)
Castorseed/Maize	0.262	0.290	3.688
Castorseed/Soyabean	0.408	0.196	2.849
Maize/Soyabean	0.665	0.184	5.717

Source: Author's estimation based on secondary data from the National Commodity and Derivative Exchange (NCDEX), Mumbai, and the Multi-Commodity Exchange (MCX), Mumbai.

Notes: All variables are the first difference of the log of futures prices. First difference of the log of futures prices are futures returns. "Period of analysis: May 2009 – August 2014 for all crops. Optimal lags are taken for all crops.

We have deflated the data with wholesale price index. We have used the seasonal dummies to remove the impact of seasonality from the spot and futures prices.

Mean equation is VAR model. Calculations based on VAR-BEKK model.

Table 9. Mean model estimates of weekly spot and futures returns in agricultural commodity markets, 2009-2014

Variable	Barley	Castor Seed	Chilli	Coriander	Cotton	Gram	Maize	Mustard	Pepper	Soya bean	Sugar	Wheat	Zeera
Mean Model (Spot Price)													
SP{1}	0.176* (0.086)	-0.155*** (0.060)	0.087 (0.067)	-0.007 (0.056)	0.004 (0.058)	-0.063 (0.083)	0.117* (0.069)	0.156 (0.113)	-0.148 (0.100)	-0.055 (0.083)	0.029 (0.071)	0.102* (0.059)	0.038 (0.058)
SP{2}			0.097 (0.059)						0.072 (0.134)		-0.100 (0.063)		0.042 (0.100)
FP{1}	-0.049 (0.040)	0.091** (0.041)	0.073 (0.046)	0.093 (0.057)	0.008 (0.055)	0.052 (0.094)	0.129** (0.053)	-0.206* (0.056)	0.228 (0.176)	0.181** (0.074)	0.205*** (0.048)	0.027 (0.073)	0.028 (0.028)
FP{2}			0.002 (0.047)						-0.015 (0.152)		0.025 (0.041)		0.006 (0.032)
Const	0.011 (0.08)	-0.051*** (0.019)	0.118** (0.052)	0.003 (0.040)	0.063 (0.038)	0.080*** (0.030)	0.010 (0.021)	0.041* (0.022)	0.045* (0.026)	0.031 (0.251)	-0.011 (0.010)	-0.002 (0.014)	-0.017 (0.016)
Mean Model (Futures Price)													
SP{1}	0.337** (0.022)	0.053 (0.064)	0.114 (0.136)	0.130 (0.082)	0.144 (0.065)	0.077 (0.076)	0.162 (0.104)	0.136 (0.106)	-0.192 (0.199)	-0.055 (0.083)	0.101 (0.098)	-0.018 (0.073)	-0.162 (0.144)
SP{2}			0.113 (0.116)						0.059 (0.299)		-0.051 (0.102)		-0.208 (0.134)
FP{1}	-0.099* (0.053)	-0.146** (0.058)	0.044 (0.074)	-0.050 (0.068)	0.229*** (0.056)	0.094 (0.089)	-0.015 (0.075)	-0.152 (0.095)	0.171 (0.312)	0.181** (0.074)	0.001 (0.067)	0.064 (0.050)	0.010 (0.060)
FP{2}			0.074 (0.082)						-0.056 (0.197)		-0.129* (0.075)		0.086 (0.063)
Const	-0.018 (0.039)	-0.058* (0.031)	-0.067 (0.065)	-0.021 (0.051)	0.088 (0.040)	0.035 (0.029)	0 (0.036)	0.060** (0.024)	0.075* (0.045)	0.038 (0.036)	-0.040** (0.015)	0.018 (0.019)	-0.065 (0.071)

Source: Author's estimation based on secondary data from the National Commodity and Derivative Exchange (NCDEX), Mumbai, and the Multi-Commodity Exchange (MCX), Mumbai.

Notes: All variables are the first difference of the log of prices. First difference of the log of prices are returns. SP denotes spot returns and FP denotes futures returns. "Cotton" refers to cotton seed oil cake. Period of analysis: May 2009 – August 2014 for all crops (daily data). Optimal lags are taken for all crops. $\{i\}$ denotes the lag i where $i = 1, 2, 3, 4 \dots$

We have deflated the data with wholesale price index. We have used the seasonal dummies to remove the impact of seasonality from the spot and futures prices.

***, **, and * indicate statistical significance at the 1%, 5% and 10% levels, respectively. Values in small brackets denote standard errors, Mean equation is VAR model. Estimations based on VAR-BEKK model.

Table 10. Variance model estimates of weekly spot and futures returns in agricultural commodity markets, 2009-2014

Variable	Barley	Castor seed	Chilli	Cori Ander	Cotton	Gram	Maize	Mustard	Pepper	Soya bean	Sugar	Wheat	Zeera
c_{11}	0.197*** (0.058)	0 (0.040)	0 (0.185)	0.040*** (0.093)	0.281*** (0.041)	0.440*** (0.074)	0.204 (0.166)	0.169*** (0.059)	0 (0.016)	0.364*** (0.053)	0.027 (0.021)	0.138*** (0.052)	0 (0.029)
c_{21}	0.188 (0.160)	0.373*** (0.056)	-0.423*** (0.082)	0.724*** (0.092)	0.664*** (0.040)	0.425*** (0.069)	0.062 (0.290)	0.347*** (0.049)	0.567*** (0.095)	0.274*** (0.061)	0.064*** (0.024)	-0.053 (0.065)	0.188*** (0.028)
c_{22}	0 (0.075)	0.023 (0.066)	0.092 (0.198)	0 (0.048)	0 (0.090)	0 (0.073)	0.254 (0.700)	0 (0.062)	0.567*** (0.095)	0 (0.128)	0 (0.059)	0.176*** (0.059)	0.549*** (0.088)
α_{11}	0.098 (0.139)	-0.285** (0.133)	-0.270** (0.130)	0.122* (0.072)	0.296 (0.120)	0.344* (0.201)	0.241 (0.331)	-0.645*** (0.110)	0.114 (0.541)	0.112 (0.118)	0.039 (0.134)	0.497*** (0.127)	0.087 (0.197)
α_{12}	0.428** (0.183)	0.198 (0.167)	-0.332 (0.248)	0.022 (0.078)	-0.253** (0.148)	-0.276** (0.120)	0.363 (0.420)	-0.555*** (0.111)	-0.276 (0.329)	0.250 (0.183)	-0.219** (0.111)	-0.243* (0.135)	-0.709 (0.240)
α_{21}	0.263*** (0.093)	-0.352*** (0.059)	-0.174* (0.089)	-0.194** (0.078)	-0.434*** (0.100)	-0.351** (0.160)	-0.218 (0.207)	0.371*** (0.098)	-0.384*** (0.089)	0.121 (0.145)	0.316*** (0.071)	0.051 (0.071)	0.044 (0.076)
α_{22}	0.179* (0.098)	-0.182*** (0.068)	0.366** (0.143)	0.192** (0.095)	0.081*** (0.075)	0.243* (0.126)	0.202 (0.145)	0.464*** (0.112)	0.162 (0.305)	0.025 (0.187)	0.418*** (0.081)	-0.464*** (0.107)	0.279** (0.110)
β_{11}	0.672*** (0.230)	0.206 (0.161)	-0.647*** (0.173)	0.343** (0.150)	0.796 (0.127)	0.427* (0.234)	0.280 (0.257)	0.927*** (0.101)	0.606 (0.875)	1.068*** (0.081)	0.260*** (0.044)	0.322** (0.129)	0.337 (0.332)
β_{12}	1.774*** (0.247)	0.509*** (0.193)	-1.014*** (0.286)	-0.568*** (0.201)	0.085 (0.211)	-0.057** (0.241)	1.069 (1.794)	0.887*** (0.191)	0.292 (0.731)	0.198*** (0.071)	1.451*** (0.097)	0.354 (0.266)	-0.917 (0.290)
β_{21}	-0.091 (0.203)	0.514*** (0.094)	-0.080 (0.110)	0.443*** (0.146)	-0.490*** (0.060)	-0.069 (0.138)	0.336 (0.272)	-0.152 (0.143)	0.095 (0.585)	-0.497*** (0.092)	0.494*** (0.024)	0.426*** (0.111)	-0.358*** (0.068)
β_{22}	-0.596*** (0.229)	0.503*** (0.165)	-0.350 (0.232)	0.516*** (0.192)	0.098 (0.103)	0.402 (0.252)	0.011 (0.311)	-0.683*** (0.236)	0.294 (0.537)	0.565*** (0.072)	-0.365*** (0.027)	0.574** (0.266)	0.187 (0.229)
δ_{11}	-0.0558 (0.520)	0.414*** (0.157)	0.258 (0.057)	-0.017 (0.104)	0.288 (0.225)	-0.032 (0.063)	0.464 (0.317)	0.258 (0.325)	0.333 (0.786)	-0.272** (0.112)	0.262* (0.148)	0.319 (0.482)	0.125 (0.177)

Table 10 (Continued)

Variable	Barley	Castor seed	Chilli	Cori Ander	Cotton	Gram	Maize	Mustard	Pepper	Soya Bean	Sugar	Wheat	Zeera
δ_{12}	-0.074 (0.691)	-0.414*** (0.158)	-0.206 (0.153)	0.208 (0.154)	0.256 (0.134)	-0.089 (0.111)	-0.692*** (0.258)	0.070 (0.268)	0.215 (0.463)	-0.427*** (0.142)	-0.042 (0.130)	0.083 (0.140)	-0.394 (0.624)
δ_{21}	0.010 (0.127)	-0.549*** (0.158)	0 (0.043)	0.224 (0.164)	0.157 (0.099)	0.215 (0.136)	-0.131 (0.230)	-0.125 (0.238)	-0.122 (0.287)	0.332* (0.173)	-0.010 (0.154)	-0.010 (0.087)	-0.072 (0.055)
δ_{22}	-0.023 (0.276)	0.223*** (0.072)	0.130 (0.147)	0.627*** (0.165)	0.198 (0.175)	0.272* (0.151)	0.167 (0.155)	-0.194 (0.256)	0.013 (0.131)	0.648*** (0.159)	-0.069 (0.090)	-0.162 (0.286)	0.144 (0.248)

Source: Author's estimation based on secondary data from the National Commodity and Derivative Exchange (NCDEX), Mumbai, and the Multi-Commodity Exchange (MCX), Mumbai.

Notes: All variables are the first difference of the log of prices . First difference of the log of prices are returns. SP denotes spot returns and FP denotes futures returns. "Cotton" refers to cotton seed oil cake. Period of analysis: May 2009 – August 2014 for all crops (daily data). Optimal lags are taken for all crops. Estimations based on VAR-BEKK model. We denotes spot returns as 1 and futures returns as 2.

We have deflated the data with wholesale price index. We have used the seasonal dummies to remove the impact of seasonality from the spot and futures prices.

***, **, and * indicate statistical significance at the 1%, 5% and 10% levels, respectively. Values in small brackets denote standard errors, and we have not reported constants in the table to save space.

$C = \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix}$; $A = \begin{bmatrix} \alpha_{11} & \alpha_{12} \\ \alpha_{21} & \alpha_{22} \end{bmatrix}$ $B = \begin{bmatrix} \beta_{11} & \beta_{12} \\ \beta_{21} & \beta_{22} \end{bmatrix}$; $D = \begin{bmatrix} \delta_{11} & \delta_{12} \\ \delta_{12} & \delta_{22} \end{bmatrix}$ denotes the coefficient matrices estimated from equation(6).

The coefficients α_{11} and α_{22} measure the impact of past shocks on the present volatility of the same time series variable.

α_{12} represents the impact of past shocks of first variable (spot) on the present volatility of the second(futures) variable. β_{12} and β_{21} measure the impact of previous uncertainties. The coefficients δ_{ii} and δ_{jj} denote the asymmetric impact of own past shocks on conditional variance.

δ_{ij} and δ_{ji} measure the cross market asymmetric response.

Table11. Portfolio features for weekly spot and futures returns, 2009-2014

Barley	Castorseed	Chilli	Coriander	Cotton	Gram	Maize	Mustard	Pepper	Soyabean	Sugar	Wheat	Zeera
Optimal portfolio weights (average)												
0.946	0.645	0.802	0.731	0.636	52.511	0.769	0.509	1	0.804	0.813	0.647	1
Average optimal hedge ratio												
0.271	0.488	0.290	0.434	0.39	0.354	0.204	0.742	0.491	0.684	0.343	0.196	0.223
Hedge effectiveness (%)												
23.521	31.026	18.177	29.365	20.589	13.383	9.439	55.648	52.269	48.457	23.930	6.070	25.567

Source: Author's estimation based on secondary data from the National Commodity and Derivative Exchange (NCDEX), Mumbai, and the Multi-Commodity Exchange (MCX), Mumbai.

Notes: All variables are the first difference of the log of prices. First difference of the log of prices are returns. Period of analysis: May 2009 – August 2014 for all crops (daily data) . Optimal lags are taken for all crops.

We have deflated the data with wholesale price index. We have used the seasonal dummies to remove the impact of seasonality from the spot and futures prices.

Mean equation is VAR model. The portfolio weights are reported for spot agricultural commodities. Calculations based on VAR-BEKK model.

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Appendix A

Role of futures markets in reducing spot price volatility

The futures markets influence spot markets by affecting the movement of agricultural commodities over time. This means that futures market have an effect on the supply and storage of the underlying agricultural commodities in spot markets. However, this can happen, if futures price performs its role of price discovery²⁷. Consequently, if growing conditions (monsoon) are good, traders on the agricultural futures markets believe that agricultural production will be higher and they expect that spot price will decrease at the time of harvest and accordingly, they amend downwards their anticipation of spot prices for upcoming months. In the new situation farmers respond by increasing their storage and not releasing agricultural commodities into the market immediately after harvest. Thus, the futures and spot markets interaction influences the flow of agricultural commodity from farmers to the market (Morgan 1999).

Futures markets through its price discovery role, provide super vision to the spot market traders regarding storage of the current crop. Thus, futures markets smoothen the flow of agricultural commodities reaching the spot market; and consequently, diminishing the chances of peaks and troughs in spot prices (Peck, 1976). The above mentioned characteristics of futures markets help us to hypothesize that futures markets reduce spot market volatility; and which in turn reduces uncertainty for the farmers. Futures markets strengthen the efficiency of stock holding, which can be regarded as Pareto improvement. Thus, futures markets may be a feasible policy option for the agricultural sector in the absence of government support policies. Several studies have found that the futures markets have reduced spot price volatility (Powers, 1970; Gilbert, 1989 and Netz 1995). However, other studies have indicated that futures markets destabilise spot price through speculation (Newbery, 1987; Driskill *et al.* 1991).

²⁷ Price discovery here means that futures price is a good forecaster of spot price which takes into consideration storage costs, transportation costs, etc.

Appendix B

Optimal hedge ratio (OHR) determination

Suppose a mill owner wants to protect themselves from agricultural commodity price fluctuations. The return on the mill owner portfolio of futures spot positions can be written as:

$$Ret_{mt} = Ret_{spt} - \beta_t Ret_{fpt}$$

where Ret_{mt} is the return of the mill owner's portfolio from t-1 to t, Ret_{fpt} and Ret_{spt} are the returns on holding futures contracts and physical commodities respectively, β_t is the hedge ratio. β_t is defined as the number of futures contract that the mill owner should sell for every unit of physical commodity on which hedging is to be done.

The variance of the portfolio conditioned on the information available at time t-1 can be written as:

$$var(Ret_{mt} | I_{t-1}) = var(Ret_{spt} - \beta_t Ret_{fpt} | I_{t-1})$$

where I_{t-1} is the information available at time t-1.

using the rule of variance operator, we can write this as:

$$\begin{aligned} var(Ret_{mt} | I_{t-1}) &= Var(Ret_{spt} | I_{t-1}) + \beta_t^2 var(Ret_{fpt} | I_{t-1}) \\ &\quad - 2 \beta_t cov(Ret_{spt}, Ret_{fpt} | I_{t-1}) \end{aligned} \quad (17)$$

where $Var(Ret_{spt} | I_{t-1})$, $Var(Ret_{fpt} | I_{t-1})$ and $Cov(Ret_{spt}, Ret_{fpt} | I_{t-1})$ are the conditional variance of spot, conditional variance of futures, and covariance of the spot and futures returns of the agricultural commodities respectively.

We define OHR as the value of β_t which minimises the conditional variance (risk) of the hedged portfolio returns. (Chang et al 2011). We minimise the $Var(Ret_{mt} | I_{t-1})$.

we take the partial derivative of equation (17) and equate it to zero. We get the following results:

$$\beta_t^* | I_{t-1} = \frac{Cov(Ret_{spt}, Ret_{fpt} | I_{t-1})}{Var(Ret_{fpt} | I_{t-1})}$$

This can be written as:

$$\beta_t^* | I_{t-1} = \frac{h_{spfpt,t}}{h_{fp,t}}$$

where β_t is the risk minimising hedge ratio for the portfolio, $h_{spfpt,t}$ is the conditional covariance between spot and futures returns of the agricultural commodities, $h_{fp,t}$ is the conditional variance of futures returns.

Appendix C

Table C 1: Multivariate Diagnostic tests for standardized residuals (weekly), 2009-2014

Variable	Barley	Castorseed	Chilli	Coriander	Cotton	Gram	Maize	Mustard	Pepper	Soyabean	Sugar	Wheat	Zeera
Multivariate test for model adequacy													
LB-Q(5)	11.885	15.006	14.914	19.146	14.383	21.559	50.194	18.781	7.672	22.940	8.444	16.287	15.257
Prob	0.919	0.776	0.781	0.512	0.810	0.364	0.723	0.536	0.993	0.291	0.988	0.698	0.761
Test for Multivariate ARCH													
Statistic	12.17	39.43	12.30	40.30	54.42	30.81	25.37	49.76	58.49	12.71	44.39	46.51	29.36
Prob	1.00	0.706	1.00	0.670	0.158	0.947	0.991	0.289	0.085	1.000	49.72	0.410	0.965

Source: Author's estimation based on secondary data from the National Commodity and Derivative Exchange (NCDEX), Mumbai, and the Multi-Commodity Exchange (MCX), Mumbai.

Notes: All variables are the first difference of the log of prices. First difference of the log of prices are returns. Period of analysis: May 2009 – August 2014 for all crops (daily data). Optimal lags are taken for all crops.

We have deflated the data with wholesale price index. We have used the seasonal dummies to remove the impact of seasonality from the spot and futures prices.

Diagnostic tests based on VAR-BEKK model.

LB-Q(5) denotes for the Ljung-Box Q-statistic (multivariate generalizations) for standardised residuals up to 5 lags.

In LB-(Q) test, the null hypothesis is that the model is adequate or no serial correlation.

Multivariate ARCH test is used to checks for residual cross-variable ARCH.

Table C 2. Mean and variance model estimates of weekly rabi futures returns, 2009-2014

	Barley	Wheat	Coriander	Gram	Mustard
Barley lag1	-0.021 (0.070)	0.162*** (0.032)	0.016 (0.088)	-0.002 (0.050)	0.021 (0.045)
Wheat lag1	0.008 (0.068)	0.023 (0.044)	-0.156 (0.131)	0.023 (0.075)	-0.056 (0.090)
Coriander lag1	-0.063 (0.039)	-0.031** (0.012)	-0.099* (0.056)	-0.023 (0.029)	-0.036 (0.023)
Gram lag1	-0.011 (0.052)	-0.064*** (0.007)	-0.275** (0.124)	0.061 (0.058)	-0.089 (0.054)
Mustard lag1	-0.013 (0.149)	-0.064 (0.044)	-0.107 (0.135)	-0.152* (0.088)	-0.234*** (0.058)
Constant	-0.016 (0.042)	0.009 (0.018)	-0.087 (0.074)	-0.033 (0.026)	-0.015 (0.015)
Variance Model					
	Barley ($i = 1$)	Wheat ($i = 2$)	Coriander ($i = 3$)	Gram ($i = 4$)	Mustard ($i = 5$)
α_{i1}	-0.114 (0.111)	0.272* (0.154)	-0.046 (0.072)	-0.063 (0.080)	0.026 (0.093)
α_{i2}	-0.136*** (0.048)	0.508*** (0.082)	0.087** (0.040)	0.025 (0.814)	0.046 (0.056)
α_{i3}	-0.285** (0.136)	-0.283 (0.193)	0.012 (0.073)	0.184 (0.231)	-0.221 (0.481)
α_{i4}	-0.001 (0.987)	-0.150* (0.088)	-0.177*** (0.046)	0.078 (0.104)	-0.090 (0.143)
α_{i5}	-0.109* (0.149)	-0.201*** (0.287)	0.025 (0.125)	0.188* (0.259)	0.183* (0.137)
β_{i1}	0.316** (0.149)	0.438 (0.105)	-0.262* (0.113)	0.291 (0.062)	0.695*** (0.287)
β_{i2}	-0.269 (0.124)	0.100 (0.024)	0.149 (0.125)	0.207*** (0.267)	0.265 (0.317)
β_{i3}	-0.568 (0.469)	-0.897** (0.024)	-0.537*** (0.125)	0.359 (0.267)	0.752** (0.317)
β_{i4}	-0.231 (0.146)	-0.388* (0.221)	0.175** (0.088)	-0.509*** (0.085)	1.025*** (0.126)
β_{i5}	-0.392*** (0.108)	0.582*** (0.204)	-0.066 (0.087)	-0.452*** (0.174)	0.757*** (0.078)
δ_{i1}	0.296** (0.127)	0.027 (0.179)	0.042 (0.089)	-0.193* (0.112)	-0.146 (0.145)

Table C2: variance model (Continued)

	Barley ($i = 1$)	Wheat ($i = 2$)	Coriander ($i = 3$)	Gram ($i = 4$)	Mustard ($i = 6$)
δ_{i2}	-0.047 (0.050)	-0.304 (0.171)	0.082** (0.034)	-0.001 (0.065)	-0.013 (0.923)
δ_{i3}	-0.319* (0.191)	1.460*** (0.461)	-0.085 (0.143)	0.019 (0.439)	-0.642 (0.428)
δ_{i4}	0.053 (0.111)	0.336* (0.198)	-0.154** (0.065)	-0.155 (0.161)	-0.043 (0.200)
δ_{i5}	0.086 (0.099)	0.181 (0.201)	-0.261*** (0.062)	0.364** (0.158)	-0.181 (0.206)
LB-Q(5)	133.771				
Prob	0.279				
Multivariate ARCH statistic	227.10				
Prob	0.448				

Source: Author's estimation based on secondary data from the National Commodity and Derivative Exchange (NCDEX), Mumbai, and the Multi-Commodity Exchange (MCX), Mumbai.

Notes: All variables are the first difference of the log of futures prices. First difference of the log of futures prices are futures returns. Period of analysis: May 2009 – August 2014 for all crops(daily data). We denote barley, wheat, coriander, gram,zeera and mustard as 1, 2,3,4,5 and 6 respectively. Optimal lags are taken according to information criteria.

We have deflated the data with wholesale price index. We have used the seasonal dummies to remove the impact of seasonality from the spot and futures prices.

***, **, and * indicate statistical significance at the 1%, 5% and 10% levels, respectively. Mean equation is VAR model. Estimations based on VAR-BEKK model.

The coefficients α_{11} and α_{22} measure the impact of past shocks on the present volatility of the same time series variable. α_{12} represents the impact of past shocks of first variable on the present volatility of the second variable. β_{12} and β_{21} measure the impact of previous uncertainties.

The coefficients δ_{ii} and δ_{jj} denote the asymmetric impact of own past shocks on conditional variance.

δ_{ij} and δ_{ji} measure the cross market asymmetric response. LB-Q(5) denotes for the Ljung-Box Q-statistic (multivariate generalizations) for standardised residuals up to 5 lags. In LB-(Q) test, the null hypothesis is that the model is adequate or no serial correlation.

Multivariate ARCH test is used to check for residual cross-variable ARCH.

Table C 3. Mean and variance model estimates of weekly kharif futures returns, 2009-2014

Mean Model			
	Castorseed	Maize	Soyabean
Castorseed lag1	0.020 (0.075)	-0.012 (0.034)	0.044 (0.044)
Maize lag1	-0.048 (0.124)	0.0107 (0.065)	-0.011 (0.083)
Soyabean lag1	0.032 (0.056)	0.053 (0.044)	-0.007 (0.079)
Constant	0 (0.045)	-0.021 (0.035)	0.044 (0.040)
Variance Model			
	Castorseed(i=1)	Maize(i=2)	Soyabean(3)
α_{i1}	0.241 (0.164)	0.435*** (139)	0.059 (0.132)
α_{i2}	0.102 (0.112)	0.287** (0.141)	0.055 (0.088)
α_{i3}	0.087 (0.196)	-0.186 (0.159)	0.421*** (0.090)
β_{i1}	0.766*** (0.095)	-0.021 (0.201)	-0.320*** (0.088)
β_{i2}	-0.071 (0.062)	0.795*** (0.078)	-0.053 (0.049)
β_{i3}	-0.023 (0.096)	-0.017 (0.077)	0.746*** (0.103)
δ_{i1}	-0.275 (0.288)	-0.234 (0.360)	-0.278* (0.167)
δ_{i2}	-0.270* (0.156)	0.194 (0.149)	0.114 (0.084)
δ_{i3}	-0.335** (0.155)	-0.402*** (0.153)	0.392*** (0.199)
LB-Q(5)	40.704		
Prob	0.654		
Multivariate ARCH statistic	164.07		
Prob	0.796		

Source: Author's estimation based on secondary data

Notes: All variables are the first difference of the log of futures prices (returns). We denote castorseed, maize, and soyabean as 1, 2 and 3 respectively. ***, **, and * indicate statistical significance at the 1%, 5% and 10% levels, respectively. Mean equation is VAR model.

The coefficients α_{11} and α_{22} measure the impact of past shocks on the present volatility of the same time series variable. α_{12} represents the impact of past shocks of first variable on the present volatility of the second variable. β_{12} and β_{21} measure the impact of previous uncertainties. Estimations based on VAR-BEKK model. The coefficients δ_{ii} and δ_{jj} denote the asymmetric impact of own past shocks on conditional variance.

δ_{ij} and δ_{ji} measure the cross market asymmetric response. LB-Q(5) denotes for the Ljung-Box Q-statistic (multivariate generalizations) for standardised residuals up to 5 lags. In LB-(Q) test, the null hypothesis is that the model is adequate or no serial correlation.

Multivariate ARCH test is used to check for residual cross-variable ARCH.

Table C4. Portfolio features for weekly rabi crops futures returns, 2009-2014

Crops combination	Optimal portfolio weights (average)	Average optimal hedge ratio	Hedge effectiveness (%)
Barley/Wheat	0.231	0.370	5.581
Barley/Coriander	0.772	0.115	3.532
Barley/Gram	0.467	0.136	1.666
Barley/Coriander	0.370	0.289	5.622
Wheat/Coriander	0.874	0.009	0.065
Wheat/Gram	0.728	0.131	3.774
Wheat/Mustard	0.641	0.110	2.006
Coriander/Gram	16.376	0.516	8.919
Coriander/Mustard	11.616	0.548	7.538
Gram/Mustard	0.372	0.511	19.641

Source: Author's estimation based on secondary data from the National Commodity and Derivative Exchange (NCDEX), Mumbai, and the Multi-Commodity Exchange (MCX), Mumbai.

Notes: All variables are the first difference of the log of futures prices. First difference of the log of futures prices are futures returns. "Period of analysis: May 2009 – August 2014 for all crops(daily data). Optimal lags are taken for all crops.

We have deflated the data with wholesale price index. We have used the seasonal dummies to remove the impact of seasonality from the spot and futures prices.

Mean equation is VAR model. Calculations based on VAR-BEKK model.

Table C5. Portfolio features for weekly kharif crops futures markets

	Optimal portfolio weights (average)	Average optimal hedge ratio	Hedge effectiveness (%)
Castorseed/Maize	0.301	0.058	0.150
Castorseed/Soyabean	0.436	0.124	1.239
Maize/Soyabean	0.717	0.273	13.398

Source: Author's estimation based on secondary data from the National Commodity and Derivative Exchange (NCDEX), Mumbai, and the Multi-Commodity Exchange (MCX), Mumbai.

Notes: All variables are the first difference of the log of futures prices. First difference of the log of futures prices are futures returns. "Period of analysis: May 2009 – August 2014 for all crops. Optimal lags are taken for all crops.

We have deflated the data with wholesale price index. We have used the seasonal dummies to remove the impact of seasonality from the spot and futures prices.

Mean equation is VAR model. Calculations based on VAR-BEKK mode